

Lab 08 – MATH 240 – Computational Statistics

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Abstract

This lab aims to analyze data of favorable sentencing from federal judges to decide if racial discrimination more closely resembled a normal distribution or a Pareto distribution. Ultimately, we found that the data fit a normal distribution better than a Pareto distribution, indicating that racially discriminatory decisions were not present in only a select few judges. We were able to find this out by first estimating the parameters for the distributions by guessing, then by calculations.

Keywords: Normal distribution; Pareto distribution; `ggplot()`; Method of Moments; Maximum Likelihood Estimator; parameters

1 Introduction

In this lab, we analyze if anti-Black discrimination occurs amongst a few judges or more generally over many judges. Some researchers have argued that racial discrimination amongst federal judges fits a Pareto distribution, meaning there is a measurable amount of discrimination that's higher in a select few judges rather than in many judges. Other researchers argue that racial discrimination fits best under a normal distribution, indicating that anti-Black judicial decisions occurred under roughly normal estimates across many federal judges.

This lab also explores the method to finding the mathematically correct parameters to accurately map a normal and Pareto distribution onto data. We used the Method of Moments (MOM) or the Maximum Likelihood Estimator (MLE) processes to find exact parameters.

2 Methods

The data for this lab contained 239,310 federal cases during a specific time frame. It is displayed as a .csv that we can read and clean using `tidyverse` (3). We made histograms, normal distributions, and Pareto distributions using `ggplot()` within `tidyverse`. To make a Pareto distribution in R, we used the `VGAM` package (4), allowing us to use the Pareto functions. To find the most accurate parameters, we used the `nleqslv` package (1), which helps us solve the equations for the moments of the normal and Pareto MOMs.

2.1 Cleaning the Data

After reading in our data and storing it as `judgedat`, we cleaned the data from 239,310 cases down to 142,609 cases that had White or Black defendants that weren't a minor. In order to do the cleaning, we made a subset of the data to include only the rows where the defendant was at least 18 years old, and then we subsetting that data to include only the rows where the race was either "White" or "Black".

2.2 Summarizing and Graphing the Data

We created a new data set called `judgesummary`, where we grouped the data by each judge and race so that we could summarize it. We asked R for the number of cases each judge presided over, as well as the proportion of times they gave a sentence "Below Range", "Within Range", and "Above Range", according to the Booker guidelines. We used the `pivot_wider()` function to take the values from the summary and the names from the race column. We also used the `mutate()` function to make a new column that subtracted the proportion of favorable sentences below the guidelines (White-Black).

After summarizing the data so that we could properly display it, we used the `ggplot()` function to create a histogram from the column we created. The x axis was the difference in proportion of favorable sentences (White-Black) and the y axis was the density.

2.3 Distribution of the Data - Guessing

We first used a normal distribution to map over our histogram. To create the distribution, we made a tibble and created new columns to represent the Probability Density Function (PDF), since we were working with density in our histogram. To make the PDF, we used the `dnorm()` function, since "d" is the prefix used to make a PDF normal distribution. We plugged this data into a `ggplot()` to make the curve, and used guess-and-check to determine the mean(μ) and standard deviation(σ) parameters. To make the curve and the histogram appear together, we pasted the normal curve code into the same plot as our histogram code.

To make a Pareto distribution, we followed many of the same steps with slight changes to our code. To start, we had to use the `VGAM` package, since base R cannot create a Pareto distribution. We made another tibble saved as `plot.data2`,

and added another column to represent the PDF. We used the `dpareto` function, with “d” as the prefix again. We plugged this data into another `ggplot()` to make the curve, and did guess-and-check to determine the Pareto parameters, `scale(beta)` and `shape(alpha)`. To make the curve and histogram appear together, we pasted the Pareto curve code with our histogram code.

2.4 Distribution of the Data - Calculations

In order to find the exact parameters for our normal and Pareto distributions, we can use the MOM or MLE processes. For the MOM process for a normal distribution, the moments functions of μ and σ . Ultimately, we wanted to set these moments equal to the sample mean, $\bar{x}$ and $\bar{x^2}$, respectively. We did this by creating a function of the parameters and data, with μ and σ as a parameter. We then created an object for the difference between the first moment and ($\text{mean}(\bar{x})$) and the second moment and ($\text{mean}(\bar{x^2})$), aiming for the differences to be as close to 0 as possible. We were able to use the `nleqslv` package to evaluate these parameters for us. After getting the exact parameters, we then simply repasted our normal distribution data along with our normal curve and histogram code, but this time, with the parameters we calculated, rather than the ones we guessed.

We can also use the MLE process to find the exact parameters for our normal distribution. We created a function of the parameters and data and created a logarithmic function of the PDF. We then used the `optim` function to find the maximum of the function we created, making note to ask for the maximum of the negative logarithmic function (because R automatically will find the minimum of the function as its optimization). Finding the minimum of the negative function is really the maximum of the positive function. After evaluating the exact parameters, we are able to plug the μ and the σ in for the mean and standard deviation of our normal distribution, and paste that onto our histogram.

For the MOM process for a Pareto distribution, the code is very similar. The only things that are changed are the moments used in the function as well as the parameters, which are α and β . Additionally, when calculating the differences between the moments and the sample means, we included a penalty for when α is less than 1 and 2, respectively, because that is a condition of the Pareto moments.

For the MLE process, the code is extremely similar to that of a MLE for a normal distribution, except we had to filter our data to only include the positive differences in proportions. This is because a Pareto distribution can only plot positive values, and we had some negative proportions in our data. After getting our parameters from the MOM and MLE processes, we could replot the Pareto distribution onto our histogram using the more accurate values.

3 Results

After graphing both a normal and Pareto distribution onto our histogram, it's clear that the normal curve is a better fit for this data.

Difference in Proportion of Favorable Sentences Between

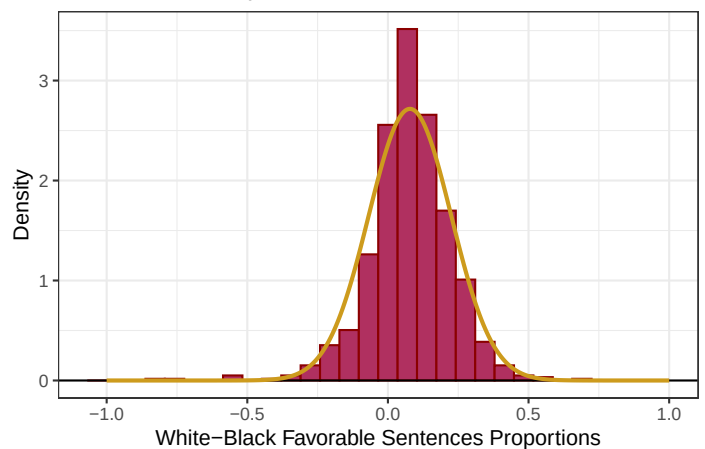


Figure 1: A histogram with the exact MLE parameters for the normal curve.

Difference in Proportion of Favorable Sentences Between

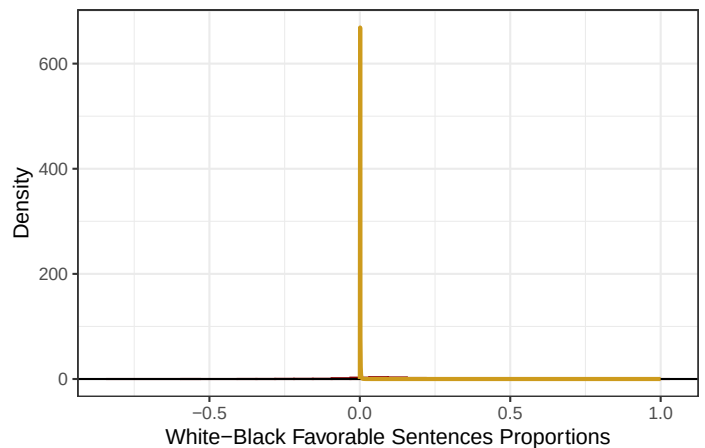


Figure 2: A histogram with the exact MLE parameters for the Pareto curve (the data is squished to the bottom due to the large peak of the Pareto distribution).

4 Discussion

Ultimately, since a normal distribution better fits this data, we can conclude that racial discrimination amongst a select few judges cannot be measured with this data. Instead, the discrimination is normally distributed amongst a wide array of judges. The data does show slight racial discrimination, as the mean difference in proportion is above 0, indicating that on average, a judge will do a favorable sentence more often on White defendants than Black defendants.

References

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